1. Subgroup traps in lift-dense factor bases

2026-11-07. A follow-up to the C1 toy ladder (PR #2704) and the lift-dense study (PR #2686). The protocol was frozen before any computation at `7ac2099b8`, and Amendment 1 at `fe8b2b62e`. The run is `reports/run-20261107T2/`. Every statement is an unverified assertion of the producing session [pcc:unverified].

1.1. Question

In the C1 ladder (field $\mathbb{F}_{Q^4}$, $Q = 2^k$, $n = 4$), the positive control A0, $y^2 + xy = x^3 + 1$, returned zero decompositions on every sampled target at $k = 7$ and 9, where the counting identity predicts about 15.5 per target. At $k = 5$ its counts were extremely clustered. The registered checks H-Y1 and H-Y3 failed on A0.

1.2. Mechanism

T1, subfield trap. Let $K = \mathbb{F}_2(u\mathbb{F}_Q, a, b)$. A lifting point with $x \in K$ has $y \in K_2 \cap \mathbb{F}_{2^N}$, where K_2 is the quadratic extension of K. So the factor base lies in $E(K^*)$ with $K^* = K_2 \cap \mathbb{F}_{2^N}$. Every sum of factor-base points stays in that subgroup. **A trap occurs if and only if $[K : \mathbb{F}_2] < N$ and $2[K : \mathbb{F}_2] \neq N$** (Amendment 1).

On A0′s standard line, $K = \mathbb{F}_Q$. A point with $x \in \mathbb{F}_Q$ satisfies $\sigma_Q P = \pm P$, so the factor base lies in $H = E(\mathbb{F}_Q) + \ker(\sigma_Q + 1)$. The two pieces meet only in $E(\mathbb{F}_Q)[2]$, which has order 2, so $|H| = |E(\mathbb{F}_{Q^2})|\ /2$.

T2, 2E concentration. $P \in 2E$ if and only if $\mathrm{Tr}(x_P) = \mathrm{Tr}(a)$. Every C1 factor base has $\mathrm{Tr}(x) = 0$. A sum of four points therefore lies in $2E$, or in the matching coset when $\mathrm{Tr}(a) = 1$.

1.3. Results (registered)

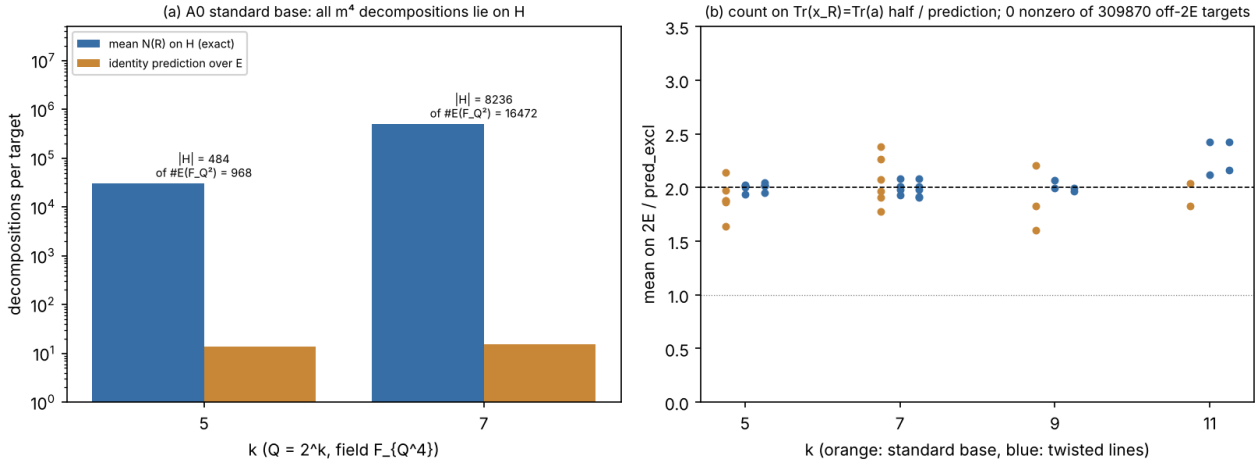


Figure 1: (a) A0 standard base: the exact mean count on H against the identity prediction over E. (b) Every frozen row: on targets with $\mathrm{Tr}(x_R) = \mathrm{Tr}(a)$, the mean count divided by the prediction. Targets off $2E$ all have count 0.

- **Identity.** A second full run (`run-20261107T1`) agrees on every frozen field. Regenerating the targets with the run's seeds reproduces every frozen count: 72 rows, 0 mismatches.
- **P1 PASS.** At $k = 5, 7, 9$, every factor-base point of A0 `std` and `tw_a` ($[K : \mathbb{F}_2] = k$, trapped) lies in H.
  - $k = 5$: 16 of the 20,000 targets lie in $E(\mathbb{F}_{Q^2})$, 9 of them in H. No target outside H has a nonzero count.
  - $k = 7$ and $k = 9$: none of the 8,000 and 1,500 targets lies in $E(\mathbb{F}_{Q^2})$, and every count is 0.
  - A target lands in H with probability $|H|\ /\ |E(\mathbb{F}_{Q^4})| \approx 1/(2Q^2)$: 0.25 expected hits at $k = 7$, 0.003 at $k = 9$.
- **P2 PASS.** $\sum_{R \in E(\mathbb{F}_{Q^2})} N(R) = m^4$ exactly:
  - $k = 5$: 14,776,336 over 968 points;
  - $k = 7$: 4,162,314,256 over 16,472 points.

  The support is exactly H: 484 and 8,236 points, half of $E(\mathbb{F}_{Q^2})$ in each case, and every point of H is reached. The mean on H is $m^4/\ |H|$: 30,530 and 505,381.
- **P3 PASS.** Across all 72 rows (every curve and arm at $k = 5$ to 11), the 309,870 (row, target) pairs with $\mathrm{Tr}(x_R) \neq \mathrm{Tr}(a)$ have zero decompositions. On the other half, the mean count divided by the prediction has a median of 2.007 over the non-A0 rows (range 1.60 to 2.43).
- **P4 PASS.** No line that the merged studies call applicable is trapped:
  - Oakley groups 3 and 4: $[K : \mathbb{F}_2] = N$; for odd N the trap is impossible.

- c2pnb176v1 to 368w1: every recorded line has $[K : \mathbb{F}_2] = N$. Since $\deg b = 16$ on every curve, no $u \in \mathbb{F}_{2^{16}}$ can trap.
- C1 A1, A2 and B1 to B3: 54 rows, none trapped.

1.4. Post hoc: the borderline case is not benign

Amendment 1 called the case $2[K : \mathbb{F}_2] = N$ benign: the factor base lies in $E(K) + \ker(\sigma_K + 1)$, which equals $2E$. A0 `tw_b` ($u \in \mathbb{F}_4 \setminus \mathbb{F}_2$, $[K : \mathbb{F}_2] = 2k$) contradicts that in the sense that matters.

Write the factor base as $A \cup B$, with $A \subset E(K)$ and $B \subset \ker(\sigma_K + 1)$. Each piece is a group of size about Q^2, but a single line point covers only about Q of it. A 4-sum therefore reaches a generic point of $2E$ only when exactly two points come from each piece. The typical target, and every cryptographic target of odd order, sees only the fraction $\binom{4}{2}p^2(1-p)^2$ of the mass, where $p = |B|\ /m$. The rest sits on sets of relative size $O(1/Q)$.

Fraction of the 2E-half targets with zero count. A0 `tw_b` (borderline): 74 %, 72 % and 74 % at $k = 5, 7, 9$. A1 `tw_a` (full degree): 31 %, 27 % and 25 %. The predicted generic share for `tw_b`, $6p^2(1-p)^2$, is 0.29, 0.37 and 0.37.

This is algebra written after the data, with descriptive statistics. It is not a registered test. No applicable line in #2686 or #2704 is borderline (P4: all have $[K : \mathbb{F}_2] = N$). The construction rule becomes: **use a line only when $[K : \mathbb{F}_2] = N$.**

1.5. What it means

- A0 was never a valid positive control. Its H-Y1 and H-Y3 failures in #2704 are explained exactly, and the verdicts stay as registered.
- The yield and cost-model rows of #2686 and #2704 stand (P4).
- The factor 2 on $2E$ is accounting. Cryptographic targets of odd order lie in $2E$.
  - For even n the standard base is in $2E$ too, so the twisted/standard ratio is unchanged.
  - For odd n (Oakley, $n = 5$), the ratio against the plain $\mathbb{F}_q$ base becomes 2^{n+1}. This is derived, not measured. The extra 2 is the known cofactor mechanism, available to any trace-zero base (`LEAD-ICB-cofactor-torsion-quotients`), so it is not an improvement.
- ECC2K-130: $N = 131$ is prime, so $K \in \{\mathbb{F}_2, \mathbb{F}_{2^{131}}\}$ and no subfield trap is possible.
- This complements `lift_dense_torsion_pairing_20261106`, which gives the rule to skip $u \in b^{1/4}\mathbb{F}_q$ and wrote the subfield explanation as unmeasured algebra for $n = 2$. A0′s line, $b = 1$, belongs to both classes.

1.6. Not claimed

No ECDLP cost, and no leaderboard, benchmark or rho row. Execution was local: these are exact algebraic checks of frozen data (AGENTS.md §9a, as for the parent studies). The KB was unavailable (no `research-kb` binary, MCP not connected). The lead atlas and `rg` were used instead.